



线性代数

一个大题 两个小题

① 12' 化二次型为标准形或规范形

② 5' 合同

③ 5' 正定=次型

$$f(x_1, x_2, x_3) = (x_1, x_2, x_3) \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

二. 合同变换、二次型的合同标准形、规范形

1. 线性变换

对于 n 元二次型 $f(x_1, x_2, \dots, x_n)$, 若令

$$\begin{cases} x_1 = c_{11}y_1 + c_{12}y_2 + \dots + c_{1n}y_n, \\ x_2 = c_{21}y_1 + c_{22}y_2 + \dots + c_{2n}y_n, \\ \dots\dots\dots \\ x_n = c_{n1}y_1 + c_{n2}y_2 + \dots + c_{nn}y_n, \end{cases} \quad (*)$$

记 $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, $C = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \dots & c_{nn} \end{bmatrix}$, $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$, 则(*)式可写为

$$x = Cy,$$

其中(*)式称为从 $y_1, y_2, \dots, y_n$ 到 $x_1, x_2, \dots, x_n$ 的线性变换. 若线性变换的系数矩阵 C 可逆, 即 $|C| \neq 0$, 则称为可逆线性变换. 现给出 $f(x) = x^T A x$, 令 $x = Cy$, 则

$$f(x) = (Cy)^T A (Cy) = y^T (C^T A C) y.$$

记 $B = C^T A C$, 则

$$f(x) = y^T B y = g(y).$$

此时, 二次型 $f(x) = x^T A x$ 通过线性变换 $x = Cy$ 得到一个新二次型 $g(y) = y^T B y$.

★ 二次型的标准形、规范形

设有交叉项

系数仅为 1, -1, 0

例 6.2

用正交变换化二次型

$$f(x_1, x_2, x_3) = 2x_1^2 + 5x_2^2 + 5x_3^2 + 4x_1x_2 - 4x_1x_3 - 8x_2x_3$$

为标准形, 并求所作的正交变换

① $A = \begin{bmatrix} 2 & 2 & -2 \\ 2 & 5 & -4 \\ -2 & -4 & 5 \end{bmatrix}$

② $|\lambda E - A| = \begin{vmatrix} \lambda-2 & -2 & 2 \\ -2 & \lambda-5 & 4 \\ 2 & 4 & \lambda-5 \end{vmatrix} = \begin{vmatrix} \lambda-2 & -2 & 2 \\ -2 & \lambda-5 & 4 \\ 0 & \lambda-1 & \lambda-1 \end{vmatrix} = (\lambda-1)(-1) \begin{vmatrix} \lambda-2 & 2 \\ -2 & 4 \end{vmatrix} + (\lambda-1) \begin{vmatrix} \lambda-2 & -2 \\ -2 & \lambda-5 \end{vmatrix}$
 $= (\lambda-1) [\lambda^2 - 7\lambda + 10 - 4 - 4 - 4\lambda + 8]$
 $= (\lambda-1) (\lambda^2 - 11\lambda + 10) = (\lambda-1)^2 (\lambda-10) = 0$
 $\lambda_1 = \lambda_2 = 1 \quad \lambda_3 = 10$

③ 当 $\lambda=1$ 时 $\lambda E - A = \begin{bmatrix} -1 & -2 & 2 \\ -2 & -4 & 4 \\ 2 & 4 & -4 \end{bmatrix}$ $\xi_1 = (2, 0, 1)^T$
 $\xi_2 = (-2, 1, 0)^T$

当 $\lambda=10$ 时 $\lambda E - A = \begin{bmatrix} 8 & -2 & 2 \\ -2 & 5 & 4 \\ 2 & 4 & 5 \end{bmatrix} \sim \begin{bmatrix} 8 & -2 & 2 \\ -2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ $\phi_3 = (1, 2, -2)^T$

④ $\phi_1' = \phi_1 = (2, 0, 1)^T$
 $\phi_2' = \phi_2 - \frac{\phi_2 \cdot \phi_1'}{\phi_1' \cdot \phi_1'} \phi_1'$
 $= (-2, 1, 0)^T + \frac{+4}{5} (2, 0, 1)^T$
 $= (-\frac{2}{5}, 1, \frac{4}{5})^T$
 $\begin{cases} \eta_1 = \frac{3}{\sqrt{14}} (2, 0, 1)^T \\ \eta_2 = \frac{1}{\sqrt{14}} (-2, 5, 4)^T \\ \eta_3 = \frac{1}{\sqrt{14}} (1, 2, -2)^T \end{cases}$ $Q = \frac{\sqrt{14}}{15} \begin{bmatrix} 6 & -2 & \sqrt{5} \\ 0 & 5 & 2\sqrt{5} \\ 3 & 4 & -2\sqrt{5} \end{bmatrix}$ $\Lambda = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 10 \end{bmatrix}$

⑤ 令 $X = QY \Rightarrow f = X^T A X = (QY)^T A (QY) = Y^T (Q^T A Q) Y$
 $\stackrel{\text{正交矩阵}}{=} Y^T (Q^T A Q) Y = Y^T \Lambda Y$

例 6.1

例 6.1 化二次型

$$f(x_1, x_2, x_3) = x_1^2 + 2x_1x_2 + 2x_1x_3 - x_2^2 - 2x_2x_3 - x_3^2$$

为标准形, 并写出所作的可逆线性变换. $\leftarrow$ 没说用“正交法”

将某个 x_i 的平方项与其有关的混合项 一次配成一个完全平方, 如法炮制

$$\begin{aligned} f(x_1, x_2, x_3) &= (x_1^2 + 2x_1x_2 + 2x_1x_3) - x_2^2 - 2x_2x_3 - x_3^2 \\ &= (x_1 + x_2 + x_3)^2 - x_2^2 - x_3^2 - 2x_2x_3 - x_2^2 - 2x_2x_3 - x_3^2 \\ &= (x_1 + x_2 + x_3)^2 - 2(x_2^2 + x_3^2 + 2x_2x_3) \\ &= (x_1 + x_2 + x_3)^2 - 2(x_2 + x_3)^2 + 0 \cdot x_3^2 \end{aligned}$$

n 元要 n 换, 缺项要补项

令 $\begin{cases} y_1 = x_1 + x_2 + x_3 \\ y_2 = x_2 + x_3 \\ y_3 = x_3 \end{cases} \Rightarrow f(x_1, x_2, x_3) = g(y_1, y_2, y_3) = y_1^2 - 2y_2^2$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad C_1$$

$$Y = C^{-1} X$$

$X = CY$
 $\Rightarrow C: \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & -1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \Rightarrow$
 $C = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

补充: 进一步化为规范形 并导出 D

$\begin{cases} z_1 = y_1 \\ z_2 = \sqrt{2}y_2 \\ z_3 = y_3 \end{cases} \quad h(z_1, z_2, z_3) = z_1^2 - z_2^2$ $\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1 & & \\ & \sqrt{2} & \\ & & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ $Z = C_2 Y = C_2 C_1 X$
 $X = C_1^{-1} C_2^{-1} Z$ 除非 $\lambda = \{1, -1, 0\}$

配方法配出来的系数不是特征值,

正交法不能化到规范形

三 惯性定理

无论选取什么样的可逆线性变换, 将二次型化成标准形或规范形, 其正项个数 p , 负项个数 q 都是不变的, p 称为正惯性指数, q 称为负惯性指数.

【注】(1) 若二次型的秩为 r , 则 $r = p + q$, 可逆线性变换不改变正、负惯性指数;

(2) 两个二次型(或实对称矩阵)合同的充要条件是有相同的正、负惯性指数, 或有相同的秩及正(或负)惯性指数.

前提是

例 6.4

A, B 实对称下: A, B 合同 $\Leftrightarrow p, q$ 相等

例 6.4 设 $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, 则下列矩阵中与 A 合同的是().

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$p=3, q=0$

(B) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$p=2, q=1$

(C) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$p=1, q=2$

(D) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$p=0, q=3$

A: $x_1^2 + 4x_1x_2 + x_3^2$
 $= (x_1 + 2x_2)^2 - 3x_2^2 + x_3^2$
 $p=2, q=1 \Rightarrow B$

例 6.2

没有平方, 创造平方

例 6.2 将二次型 $f(x_1, x_2, x_3) = x_1x_2 + x_1x_3 - x_2x_3$ 化为规范形, 并求所用的可逆线性变换.

$$\begin{aligned} \begin{cases} x_1 = y_1 + y_2 \\ x_2 = y_1 - y_2 \\ x_3 = y_3 \end{cases} & f(x_1, x_2, x_3) = \frac{y_1^2 - y_2^2}{2} + \frac{y_1y_3 + y_2y_3 - y_1y_3 + y_2y_3}{2} \\ & = \frac{y_1^2}{2} - \frac{(y_2 - y_3)^2}{2} + \frac{y_3^2}{2} \\ & = z_1^2 - z_2^2 - z_3^2 \\ X &= \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} Y & Z = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} Y \\ & X = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} Z \end{aligned}$$

四 正定二次型及其判别

1. 定义

n 元二次型 $f(x_1, x_2, \dots, x_n) = x^T A x$. 若对任意的 $x = [x_1, x_2, \dots, x_n]^T \neq 0$, 均有 $x^T A x > 0$, 则称 f 为正定二次型, 称二次型的对应矩阵 A 为正定矩阵.

2. 二次型正定的充要条件

n 元二次型 $f = x^T A x$ 正定 $\Leftrightarrow$ 对任意 $x \neq 0$, 有 $x^T A x > 0$ (定义)

$\Leftrightarrow f$ 的正惯性指数 $p = n$

$\Leftrightarrow$ 存在可逆矩阵 D , 使 $A = D^T D$

$\Leftrightarrow A \simeq E$

$\Leftrightarrow A$ 的特征值 $\lambda_i > 0 (i = 1, 2, \dots, n)$

$\Leftrightarrow A$ 的全部顺序主子式均大于 0. ★

【注】 设 $A = (a_{ij})_{n \times n}$, 则

$$|A_k| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{vmatrix}$$

称为 n 阶矩阵 A 的 k 阶顺序(或左上角)主子式. 当 k 取 $1, 2, \dots, n$ 时, 就得到 A 的 n 个顺序主子式.

例 6.5

例 6.5 判别二次型

$$f(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_2 + 2x_1x_3 + 2x_2x_3$$

的正定性.

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

方法一 $|2| > 0$

$$| \begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix} | > 0$$

$$| \begin{smallmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{smallmatrix} | = 8 + 1 + 1 - 2 - 2 - 2 > 0 \quad \text{正定}$$

$$\text{方法二 } |E - A| = \begin{vmatrix} \lambda - 2 & -1 & -1 \\ -1 & \lambda - 2 & -1 \\ -1 & -1 & \lambda - 2 \end{vmatrix} = (\lambda - 4) \begin{vmatrix} 1 & -1 & -1 \\ 1 & \lambda - 2 & -1 \\ 1 & -1 & \lambda - 2 \end{vmatrix} = (\lambda - 4) \begin{vmatrix} 1 & 0 & 0 \\ 1 & \lambda & 0 \\ 1 & 0 & \lambda \end{vmatrix} = (\lambda - 4)(\lambda - 1)^2$$

正定

$$\begin{aligned} \text{方法三 } f(x_1, x_2, x_3) &= 2(x_1 + \frac{x_2}{2} + \frac{x_3}{2})^2 + \frac{3}{2}x_2^2 + \frac{3}{2}x_3^2 + x_2x_3 \\ &= 2(x_1 + \frac{x_2}{2} + \frac{x_3}{2})^2 + \frac{3}{2}(x_2 + \frac{1}{3}x_3)^2 + \frac{1}{3}x_3^2 \end{aligned}$$

正定

例 6.8

例 6.8 设实二次型 $f(x_1, x_2, \dots, x_n) = x^T A x$, 其中 $A = (a_{ij})_{n \times n}$ 是实对称矩阵, 则 $f(x_1, x_2, \dots, x_n)$ 为正定二次型的充要条件是 (B).

(A) A^* 是正定矩阵

(B) A^{-1} 是正定矩阵

(C) $f(x_1, x_2, \dots, x_n)$ 的负惯性指数为零

(D) 存在 n 阶实矩阵 C , 使得 $A = C^T C$

① 先设 $A = A^T$

A 正定 $\Leftrightarrow A^T$ 正定

② 再用充要条件

A 正定 $\Leftrightarrow A^T$ 正定

A 正定 $\xrightarrow{\text{充要}} A^T$ 正定

$$A^T = (A^T)^T = (A^T)^T$$

$$\text{又: } A \text{ 的 } \lambda > 0 \quad A^T \text{ 的 } \frac{1}{\lambda} > 0$$

$$A^* = (A^T)^* = (A^T)^T = A$$

$$A \text{ 的 } \lambda > 0, \quad A^* \text{ 的 } \frac{1}{\lambda} = \frac{\lambda_2 \lambda_3 \dots \lambda_n}{\lambda_1} > 0$$

7.6.1

6.1 设矩阵

$$B = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 0 & 1 \\ 0 & 4 & 5 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

问 $f(x_1, x_2, x_3) = x^T B x$ 是否为关于 x_1, x_2, x_3 的二次型? B 是否为 f 的矩阵? 写出 f 的矩阵表示式.

不是 不是

$$\begin{aligned} f &= [x_1 \ x_2 \ x_3] \begin{bmatrix} 1 & 2 & 3 \\ -2 & 0 & 1 \\ 0 & 4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= [x_1 - 2x_2, 2x_1 + 4x_3, 3x_1 + x_2 + 5x_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= \underline{x_1^2} - \underline{2x_1x_2} + \underline{2x_1x_3} + \underline{4x_2x_3} + \underline{3x_1x_3} + \underline{x_2x_3} + \underline{5x_3^2} \\ &= 4x_1^2 + 5x_3^2 + 5x_2x_3 \\ &= \begin{bmatrix} 1 & 0 & \frac{5}{2} \\ 0 & 0 & \frac{5}{2} \\ \frac{3}{2} & \frac{5}{2} & 5 \end{bmatrix} \end{aligned}$$

7.6.2

6.2 设二次型 $f(x_1, x_2, x_3) = 2x_1^2 + 4x_2^2 + ax_3^2 + 4x_1x_2 + 4x_1x_3 + 4x_2x_3$ 的秩为 2, 则参数 $a =$

$$\begin{bmatrix} 2 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & a \end{bmatrix} \sim \begin{bmatrix} 2 & 2 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & a-2 \end{bmatrix} \Rightarrow \cancel{a=0} \quad a=2$$

7.6.3

6.3 二次型 $f(x_1, x_2, x_3) = 2x_1^2 + 2x_3^2 + 4x_1x_2 - 4x_1x_3 + 8x_2x_3$ 的规范形为().

(A) $y_1^2 + y_2^2 + y_3^2$

(B) $y_1^2 + y_2^2 - y_3^2$

(C) $y_1^2 - y_2^2 - y_3^2$

(D) $y_1^2 - y_2^2$

$$\begin{aligned} \begin{cases} x_1 = y_1 - y_2 \\ x_2 = y_1 + y_2 \\ x_3 = y_3 \end{cases} & \quad 2(y_1^2 + y_2^2 + 2y_1y_2) + 2y_3^2 + 4(y_1^2 - y_2^2) - 4(y_1y_3 - y_2y_3) + 8(y_1y_3 + y_2y_3) \\ &= 6y_1^2 - 2y_2^2 + 2y_3^2 + \underline{4y_1y_2} + \underline{12y_2y_3} + \underline{4y_1y_3} \\ &= 6(y_1^2 + \frac{2}{3}y_1y_2 + \frac{2}{3}y_1y_3) - 2y_2^2 + 2y_3^2 + 12y_2y_3 \\ &= 6(y_1 + \frac{y_2}{3} + \frac{y_3}{3})^2 - 6(\frac{2}{9}y_2y_3 - \frac{y_2^2}{9} - \frac{y_3^2}{9}) - 2y_2^2 + 2y_3^2 + 12y_2y_3 \\ &= 6(y_1 + \frac{y_2}{3} + \frac{y_3}{3})^2 - \frac{4}{3}y_2^2 + \frac{8}{3}y_3^2 + \frac{32}{3}y_2y_3 \\ &= 6(\quad)^2 - \frac{4}{3}(y_2^2 - 2y_2y_3 + 8y_2y_3) \\ &= 6(\quad)^2 - \frac{4}{3}[(y_2 + 4y_3)^2 - 18y_3^2] \\ &= 6(\quad)^2 - \frac{4}{3}(\quad)^2 + 24y_3^2 \quad (B) \end{aligned}$$

76.4

6.4 设二次型

$$f(x_1, x_2, x_3) = kx_1^2 + 2x_2^2 - 2x_3^2 + 2mx_1x_3 (m > 0),$$

其中二次型的矩阵的特征值之和为 1, 特征值之积为 -12.

(1) 求 k, m ;

(2) 用正交变换化二次型为标准形, 并求所作的正交变换及对应的正交矩阵.

$$(1) A = \begin{bmatrix} k & 0 & m \\ 0 & 2 & 0 \\ m & 0 & -2 \end{bmatrix}$$

$$|\lambda E - A| = \begin{vmatrix} \lambda - k & 0 & -m \\ 0 & \lambda - 2 & 0 \\ -m & 0 & \lambda + 2 \end{vmatrix} = (\lambda - 2) \left[(\lambda - k)(\lambda + 2) - m^2 \right]$$

$$= (\lambda - 2) \left[\lambda^2 + (2 - k)\lambda - 2k - m^2 \right]$$

$$\text{设 } \lambda_1 = 2, \lambda_2 = a, \lambda_3 = b$$

$$\begin{cases} 2 + a + b = 1 \rightarrow a + b = -1 \rightarrow a = -b - 1 \\ 2ab = -12 \rightarrow ab = -6 \quad (b+1)b = 6 \quad b^2 + b - 6 = 0 \end{cases} \begin{cases} b = -3 \quad a = 2 \\ b = 2 \quad a = -3 \end{cases}$$

$$\Rightarrow \lambda_1 = 2, \lambda_2 = 2, \lambda_3 = -3$$

$$(\lambda - 2)(\lambda + 3) = \lambda^2 + \lambda - 6 \Rightarrow \begin{cases} 2 - k = 1 \Rightarrow k = 1 \\ 2k + m^2 = 6 \Rightarrow m = 2 \end{cases}$$

$$(2) A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & -2 \end{bmatrix}$$

$$\text{当 } \lambda = 2 \text{ 时 } \lambda E - A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix} \quad \begin{cases} \xi_1 = (2, 0, 1)^T \\ \xi_2 = (0, 1, 0)^T \end{cases}$$

$$\text{当 } \lambda = -3 \text{ 时 } \lambda E - A = \begin{bmatrix} -4 & 0 & -2 \\ 0 & -5 & 0 \\ -2 & 0 & -1 \end{bmatrix} \quad \xi_3 = (1, 0, -2)^T$$

$$\begin{cases} \xi_1' = \xi_1 = (2, 0, 1)^T \\ \xi_2' = \xi_2 - \frac{\xi_2 \cdot \xi_1'}{\xi_1' \cdot \xi_1'} \xi_1' = (0, 1, 0) - \frac{0}{5} \xi_1' = (0, 1, 0)^T \\ \xi_3' = \xi_3 = (1, 0, -2)^T \end{cases}$$

$$\eta_1 = \frac{1}{\sqrt{5}} (2, 0, 1)^T$$

$$\eta_2 = \frac{1}{\sqrt{5}} (0, 1, 0)^T$$

$$\eta_3 = \frac{1}{\sqrt{5}} (1, 0, -2)^T$$

$$C = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -2 \end{bmatrix} \quad C^T A C = \Lambda = \begin{bmatrix} 2 & & \\ & 2 & \\ & & -3 \end{bmatrix}$$

$$X = CY$$

$$f = X^T A X = (CY)^T A (CY) = Y^T (C^T A C) Y = Y^T \begin{bmatrix} 2 & & \\ & 2 & \\ & & -3 \end{bmatrix} Y = 2y_1^2 + 2y_2^2 - 3y_3^2$$

7b.5

6.5 设 $f(x_1, x_2) = x_1^2 + 3x_2^2 + 4x_1x_2$, 则 $f(x_1, x_2)$ 对应的矩阵与下列矩阵不合同的是 (D).

(A) $\begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & -1 \\ -1 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$

$$f = (x_1 + 2x_2)^2 - x_2^2$$

$$x_1^2 + 2x_1x_2 + 2x_1x_2$$

7b.7

6.7 已知 $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & k+2 \end{bmatrix}$ 正定, 则 k 应满足条件 _____.

$$\begin{cases} k > 0 \\ k-1 > 0 \\ k(k+2) - (k+2) = (k-1)(k+2) > 0 \end{cases} \quad \begin{matrix} k > 1 \\ k > -2 \end{matrix}$$

7b.6

6.6 已知 $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & \frac{1}{9} \end{bmatrix}$ 合同于 $\Lambda = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$, 即存在可逆矩阵 C , 使得 $C^T A C = \Lambda$, 其中 $C =$

$$x_1^2 - 4x_2^2 + \frac{1}{9}x_3^2 = y_1^2 + y_2^2 - y_3^2$$

$$\begin{cases} x_1 = y_1 \\ x_2 = \frac{1}{2}y_2 \\ x_3 = \frac{3}{2}y_3 \end{cases} \quad C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & \frac{3}{2} & 0 \end{bmatrix}^T$$

7b.8

6.8 已知二次型 $f(x_1, x_2, x_3) = x^T A x$ 在正交变换 $x = Qy$ 下的标准形为 $y_1^2 + y_2^2$, 且 Q 的第 3 列为 $\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}^T$.

(1) 求矩阵 A ;

(2) 证明 $A+E$ 为正定矩阵, 其中 E 为 3 阶单位矩阵.

$$\therefore \Lambda = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 0 \end{bmatrix} \quad \lambda=0 \text{ 时 } \xi = \left(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2} \right)^T$$

$$X = QY \quad f = X^T A X = Y^T (Q^T A Q) Y = Y^T \Lambda Y$$

设 $[x_1, x_2, x_3]^T$ 是 $\lambda=1$ 时的特征向量

$$[x_1, x_2, x_3] \cdot \begin{bmatrix} \frac{\sqrt{2}}{2} \\ 0 \\ \frac{\sqrt{2}}{2} \end{bmatrix} = 0 \Rightarrow x_1 = -x_3$$

$$\Rightarrow \begin{cases} \xi_2 = \left(\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2} \right)^T \\ \xi_3 = (0, 1, 0)^T \end{cases}$$

$$\Rightarrow Q = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{bmatrix} \quad \Lambda = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 0 \end{bmatrix}$$

$$\begin{array}{c|c} A & A+E \\ \hline \lambda & \lambda+1 \end{array} = (2, 2, 1) > 0.$$

对称矩阵

正定

$$\Lambda = Q^T A Q = Q^T A Q$$

$$\Rightarrow A = Q \Lambda Q^T = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

300.6.1

6.1 二次型 $f(x_1, x_2, x_3) = x^T \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \\ 7 & 8 & 5 \end{bmatrix} x$ 的矩阵是

$$\begin{bmatrix} 1 & \frac{3}{2} & \frac{5}{2} \\ \frac{3}{2} & 4 & 6 \\ \frac{5}{2} & 6 & 5 \end{bmatrix}$$

300.6.2

二次型 $f(x_1, x_2, x_3) = (x_1 + x_2)^2 + (x_2 - x_3)^2 + (x_3 + x_1)^2$ 的秩为 2

$$\begin{aligned} & 2(x_1^2 + x_2^2 + x_3^2 + x_1x_2 + x_1x_3 - x_2x_3) \\ &= 2\left((x_1 + \frac{x_2}{2})^2 + \frac{3}{4}x_2^2 + \frac{3}{4}x_3^2 - x_2x_3\right) \\ &= 2\left((x_1 + \frac{x_2}{2})^2 + \frac{3}{4}(x_2^2 + x_3^2 - \frac{4}{3}x_2x_3)\right) \\ &= 2(x_1 + \frac{x_2}{2})^2 + \frac{3}{2}\left(x_2 - \frac{2}{3}x_3\right)^2 + \frac{1}{2}x_3^2 \end{aligned}$$

300.6.3

6.3 设二次型 $f(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + ax_3^2 - 4x_1x_2 - 4x_2x_3$ 经正交变换化为标准形 $f = 2y_1^2 + 5y_2^2 + by_3^2$, 则 ().

(A) $a=3, b=1$

(B) $a=3, b=-1$

(C) $a=-3, b=1$

(D) $a=-3, b=-1$

$$A = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 2 & -2 \\ 0 & -2 & a \end{bmatrix}$$

$$\begin{aligned} |\lambda E - A| &= \begin{vmatrix} \lambda-1 & 2 & 0 \\ 2 & \lambda-2 & 2 \\ 0 & 2 & \lambda-a \end{vmatrix} = (\lambda-1)[\lambda^2-3\lambda+2-4] - 2[2\lambda-2] \\ &= (\lambda-1)[\lambda^2-3\lambda-2] - 4\lambda+4 \end{aligned}$$

$$\begin{cases} (2-1)(-4)-4=0 & a-2=1 \quad (a=3) \\ (5-1)(8)-16=0 \\ (7-1)(7-3b-2)=4b-4 \end{cases}$$

$$(5-1)(8)-16=0$$

$$(7-1)(7-3b-2)=4b-4$$

$$7^3-3b^2-2b-3b^2+9b+6-4b+4=0$$

$$7^3-6b^2+3b+10=0 \rightarrow (b=-1) \quad (b)$$

300.6.4

6.4 设 A 为 3 阶实对称矩阵, 若矩阵 A 满足 $A^3 + 2A^2 - 3A = O$, 且二次型 $x^T Ax$ 的正、负惯性指数均为 1, 则二次型 $x^T Ax$ 经正交变换可化为标准形 ().

(A) $y_1^2 + 2y_2^2 - 3y_3^2$

(B) $-3y_1^2 + y_2^2$

(C) $3y_1^2 - y_2^2$

(D) $3y_1^2 - 2y_2^2 - y_3^2$

$$\cancel{A(A^2 + 2A - 3E)} \quad \lambda^3 + 2\lambda^2 - 3\lambda = 0$$

$$= A(A-E)(A+3E) \Rightarrow \lambda_1=0 \quad \lambda_2=1 \quad \lambda_3=-3$$

$$= 0 \quad 1 \quad -3 \quad (B)$$

300.6.5

6.5 设 $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, 则在实数域上与 A 合同的矩阵为 ().

(A) $\begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$

(B) $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$

$$\begin{vmatrix} \lambda-1 & -2 \\ -2 & \lambda-1 \end{vmatrix} = (\lambda-1)^2 - 4 \quad \lambda_1=3 \quad \lambda_2=-1$$

(A) $-x_1^2 - 2x_2^2 + 2x_1x_2$

(B) $2x_1^2 + 2x_2^2 - 2x_1x_2$

(C) x_1^2

(D) $x_1^2 - 4x_2^2 + x_1x_2$

$$-2(x_1 - \frac{x_2}{2})^2 + \frac{3}{2}x_2^2$$

$$2(x_1 - \frac{x_2}{2})^2 + \frac{3}{2}x_2^2$$

$$= (x_1 - 2x_2)^2 - 3x_2^2$$

$$-2x_1^2 + 2x_1x_2 - \frac{1}{2}x_2^2$$

$$x_1^2$$

300.6.6

6.6 二次型 $f(x_1, x_2, x_3) = x_1^2 + 4x_2^2 + 4x_3^2 - 4x_1x_2 + 4x_1x_3 - 8x_2x_3$ 的规范形是().

(A) $z_1^2 + z_2^2 + z_3^2$

(B) $z_1^2 - z_2^2 - z_3^2$

(C) $z_1^2 - z_2^2$

(D) z_1^2

$$(x_1 - 2x_2 + 2x_3)^2$$

(D)

300.6.7

若实对称矩阵 A 与矩阵 $B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$ 合同, 则二次型 $f = x^T A x$ 的规范形为

$$|\lambda E - B| = \begin{vmatrix} \lambda & 0 & 0 \\ 0 & \lambda - 2 & -1 \\ -1 & -1 & \lambda - 2 \end{vmatrix} = \lambda(\lambda^2 - 4\lambda + 4 - 1) = \lambda(\lambda - 1)(\lambda - 3) = 0$$

$$\lambda_1 = 0 \quad \lambda_2 = 1 \quad \lambda_3 = 3 \quad \begin{bmatrix} 0 & 1 & 3 \end{bmatrix}$$

$$f = y_2^2 + y_3^2$$

300.6.8

设二次型 $f(x_1, x_2, x_3) = x_1^2 - x_2^2 + 2ax_1x_3 + 4x_2x_3$ 的负惯性指数为 1, 求常数 a 的取值范围.

$$\begin{aligned} f &= (x_1 + ax_3)^2 - a^2x_3^2 - x_2^2 + 4x_2x_3 \\ &= (x_1 + ax_3)^2 - (x_2 - 2x_3)^2 + (4 - a^2)x_3^2 \\ 4 - a^2 &\geq 0 \quad | -a \geq 0 \\ a &\in [-2, 2] \quad a \leq 1 \\ \Rightarrow a &\in [-2, 2] \end{aligned}$$

300.6.9

6.9 用配方法将二次型 $f(x_1, x_2, x_3) = x_1^2 - 3x_3^2 - 2x_1x_2 - 2x_1x_3 - 6x_2x_3$ 化为规范形, 并写出变换矩阵.

$$\begin{aligned} f &= (x_1 - x_2 - x_3)^2 - x_2^2 - 4x_3^2 - 8x_2x_3 \\ &= (x_1 - x_2 - x_3)^2 - (x_2 + 4x_3)^2 + 12x_3^2 \end{aligned}$$

$$\begin{cases} y_1 = x_1 - x_2 - x_3 \\ y_2 = x_2 + 4x_3 \\ y_3 = 2\sqrt{3}x_3 \end{cases} \quad y = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & 2\sqrt{3} \end{bmatrix} x \quad X = CY \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & \frac{-\sqrt{3}}{2} \\ 0 & 1 & \frac{2\sqrt{3}}{3} \\ 0 & 0 & \frac{\sqrt{3}}{6} \end{bmatrix} \quad C = \begin{bmatrix} 1 & 1 & \frac{-\sqrt{3}}{2} \\ 0 & 1 & \frac{2\sqrt{3}}{3} \\ 0 & 0 & \frac{\sqrt{3}}{6} \end{bmatrix}$$

300.6.10

利用正交变换 $x = Qy$, 把二次型 $f(x_1, x_2, x_3) = x_1^2 + x_2^2 + 2x_3^2 + 2x_1x_2$ 化为标准形.

$$\begin{aligned} f(x_1, x_2, x_3) &= x_1^2 + x_2^2 + 2x_3^2 + 2x_1x_2 \\ \Rightarrow A &= \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{aligned}$$

$$|\lambda E - A| = \begin{vmatrix} \lambda - 1 & -1 & 0 \\ -1 & \lambda - 1 & 0 \\ 0 & 0 & \lambda - 2 \end{vmatrix} = (\lambda - 2)(\lambda^2 + 1 - 2\lambda - 1) = (\lambda - 2)^2 \lambda = 0$$

$$\lambda_1 = \lambda_2 = 2 \quad \lambda_3 = 0$$

$$\begin{aligned} \lambda=2 \text{ 时 } \lambda E-A &= \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{cases} \xi_1 = (0, 0, 1)^T \\ \xi_2 = (1, 1, 0)^T \\ \xi_3 = (-1, 1, 0)^T \end{cases} \Rightarrow \begin{cases} \eta_1 = (0, 0, 1)^T \\ \eta_2 = (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0)^T \\ \eta_3 = (-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0)^T \end{cases} \\ \lambda=0 \text{ 时 } \lambda E-A &= \begin{bmatrix} -1 & -1 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \end{aligned}$$

$$\text{存在 } Q = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 1 & 0 & 0 \end{bmatrix} \text{ s.t. } Q^T A Q = \begin{bmatrix} 2 & & \\ & 2 & \\ & & 0 \end{bmatrix}$$

$$\begin{aligned} f &= x^T A x = (Cy)^T A (Cy) = y^T (C^T A C) y = y^T \begin{pmatrix} 2 & & \\ & 2 & \\ & & 0 \end{pmatrix} y \\ x &= Cy \end{aligned}$$

$$= 2y_1^2 + 2y_2^2$$

300.6.11

6.11 已知 $\alpha = [1, -2, 2]^T$ 是二次型 $x^T A x = ax_1^2 + 4x_2^2 + bx_3^2 - 4x_1x_2 + 4x_1x_3 - 8x_2x_3$ 矩阵 A 的特征向量, 求正交变换化二次型为标准形, 并写出所用正交变换.

$$A = \begin{bmatrix} a & -2 & 2 \\ -2 & 4 & -4 \\ 2 & -4 & b \end{bmatrix} \quad \begin{bmatrix} a & -2 & 2 \\ -2 & 4 & -4 \\ 2 & -4 & b \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$$

$$\begin{aligned} a+4+4 &= \lambda = 9 \Rightarrow a=1 \\ -2-8-8 &= -2\lambda \Rightarrow \lambda=9 \\ 2+8+2b &= 2\lambda \Rightarrow b=4 \end{aligned}$$

$$\begin{aligned} |\lambda E - A| &= \begin{vmatrix} \lambda-a & 2 & -2 \\ 2 & \lambda-4 & 4 \\ -2 & 4 & \lambda-b \end{vmatrix} = \begin{vmatrix} \lambda-a & 2 & -2 \\ 0 & \lambda & \lambda-b+4 \\ -2 & 4 & \lambda-b \end{vmatrix} = \begin{vmatrix} \lambda-a & 2 & -4 \\ 0 & \lambda & 4-b \\ -2 & 4 & \lambda-b-4 \end{vmatrix} = \begin{vmatrix} \lambda-1 & 2 & -4 \\ 0 & \lambda & 0 \\ -2 & 4 & \lambda-8 \end{vmatrix} \\ &= \lambda [\lambda^2 - 9\lambda + 8 - 8] = \lambda^2 (\lambda - 9) = 0 \quad \lambda_1 = \lambda_2 = 0 \quad \lambda_3 = 9 \end{aligned}$$

当 $\lambda=0$ 时

$$\lambda E - A = \begin{bmatrix} -1 & 2 & -2 \\ 2 & -4 & 4 \\ -2 & 4 & -4 \end{bmatrix} \rightarrow \begin{cases} \xi_1 = (2, 1, 0)^T \\ \xi_2 = (-2, 0, 1)^T \end{cases}$$

$$\xi_1' = \xi_1 = (2, 1, 0)^T$$

$$\xi_2' = \xi_2 - \frac{\xi_2 \cdot \xi_1'}{\xi_1' \cdot \xi_1'} \xi_1' = (-2, 0, 1)^T + \frac{+4}{5} (2, 1, 0)^T = (\frac{-2}{5}, \frac{4}{5}, 1)^T$$

$$\Rightarrow \begin{cases} \eta_1 = \frac{1}{\sqrt{5}} (2, 1, 0)^T \\ \eta_2 = \frac{1}{\sqrt{5}} (-2, 4, 5)^T \\ \eta_3 = \frac{1}{9} (1, -2, 2)^T \end{cases} \Rightarrow C = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -2 & 5 \\ 1 & 4 & -1 \\ 0 & 5 & 2 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 0 & & \\ & 0 & \\ & & 9 \end{bmatrix}$$

$$\begin{aligned} f &= x^T A x = y^T (C^T A C) y = y^T \Lambda y = 9y_3^2 \\ x &= Cy \end{aligned}$$

300.6.12

6.12 设 A, B 为 n 阶正定矩阵, 下列各矩阵中不一定是正定矩阵的是 (A).

(A) AB

(B) $A+B$

(C) $A^{-1}+B^{-1}$

(D) $3A+2B$

300.6.13

6.13 下列矩阵为正定矩阵的是().

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 2 & 3 \end{bmatrix}$

(B) $\begin{bmatrix} 6 & 2 & 1 \\ 2 & 0 & 5 \\ 1 & 5 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

(D) $\begin{bmatrix} 4 & 3 & 2 \\ 3 & 4 & 1 \\ 2 & 1 & -1 \end{bmatrix}$

~~AX~~ 对称! C

300.6.14

若 $f(x_1, x_2, x_3) = 2x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 - tx_2x_3$ 是正定二次型, 则 t 的取值范围是.

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & -\frac{t}{2} \\ 0 & -\frac{t}{2} & 1 \end{bmatrix}$$

$$|A| > 0, 2-1 > 0, \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & -\frac{t}{2} \\ 0 & -\frac{t}{2} & 1 \end{vmatrix} = 2 - 1 - 2 \frac{t^2}{4} = 1 - \frac{t^2}{2} > 0$$

$$\frac{t^2}{2} < 1 \quad \underline{t^2 < 2} \quad t \in (-\sqrt{2}, \sqrt{2})$$

300.6.15

6.15 设二次型 $f(x_1, x_2, x_3) = x^T A x = x_1^2 + 5x_2^2 + x_3^2 - 4x_1x_2 + 2x_2x_3$, 则对任意 $x \neq 0$, 均有().

(A) $f(x_1, x_2, x_3) > 0$

(B) $f(x_1, x_2, x_3) \geq 0$

(C) $f(x_1, x_2, x_3) < 0$

(D) $f(x_1, x_2, x_3) \leq 0$

$$A = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$5 - 1 - 4 = 0.$$

(B)